JOSHUA M LE

Houston, Texas · joshuale30@gmail.com · mininoob.github.io

SUMMARY

Software engineer who builds real-time simulation and autonomous systems. Recent work includes a medical triage training simulator, digital twins for security testing, cyber defense for industrial control systems, and robotics and LiDAR research funded by NASA. Most at home in C++, C#, and Python.

EXPERIENCE

Software Engineer, CACI International · San Antonio, Texas — Mar 2024 – Present

- Build and maintain core features of a medical triage simulator that trains medical personnel and supports research studies.
- Fixed major bugs and sped up the simulation's heaviest code, making it 50% faster so it runs smoothly in the headset.
- Rebuilt how research telemetry gets saved, adding redundant writes so every experiment captures 100% of its data.
- Helped maintain and review the team's Docker setup and Python data ingest scripts.

Software Engineer Intern, CACI International · San Antonio, Texas — Mar 2022 – Mar 2024

- Led development of a cyber defense dashboard in React, backed by an AI model the team trained to detect and stop attacks on industrial control systems (SCADA, PLC, HMI) as they happen.
- Built a digital twin platform in C# and Unity for penetration testing. It spins up virtual copies of real hardware on virtual machines, or links to live equipment and streams its data.
- Built a Lego digital twin for STEM outreach with Port San Antonio, using photogrammetry and live PLC data.

Graduate Researcher, UTSA SAVE Lab · San Antonio, Texas — Jun 2022 – Dec 2023

- Helped build a VR cybersickness dataset by recording gaze and heart rate while participants walked a procedurally generated maze for ten minutes. A model trained on it reached over 90% accuracy on cognitive load and attention (IEEE, 2025).
- Built Project Wildfire with NASA MIRO CAMEE, a HoloLens app that combines real GIS terrain, vegetation, and weather data with a plume simulation to predict where a wildfire will spread.

Undergraduate Researcher, NASA MIRO CAMEE · San Antonio, Texas — Nov 2020 – Dec 2022

- Developed a fully autonomous mapping platform that pairs a 3D printed rover with a drone to collect LiDAR terrain data.
- Applied the V-RBNN algorithm and sensor fusion in ROS to clean LiDAR point clouds for coastal reconnaissance.

PROJECTS

Polymage · C++20, Vulkan — in progress

Building a game engine from scratch. A GPU compute shader raymarcher renders an infinite streamed voxel world, and the deterministic 60 Hz simulation (EnTT, Jolt) gets replayed by CI on every commit to catch drift.

OriginalCopy · Swift, FastAPI, Docker — shipped

Built and shipped a photobooth platform end to end. An iOS app runs the booth's camera and thermal printer, and a FastAPI backend handles device enrollment, telemetry, and owner accounts, deployed in containers with instant rollback.

Pinq · Rust, Tauri, SolidJS — github.com/Mininoob/pinq

Built a notes app for Windows and macOS that keeps data on the user's own machines. Devices sync directly with no server and end to end encryption, edits land in under a second, and an ML model on the device suggests tags.

SKILLS

Languages	C++, C, C#, Python, Rust, Swift, Kotlin, JavaScript, TypeScript
Simulation	Unity 3D, Vulkan, ROS, LiDAR, sensor fusion, digital twins, AR/VR (HoloLens), photogrammetry, MATLAB
Tools	Git, Docker, Linux, React, FastAPI, SQLite, MongoDB, MySQL, Wireshark, Ghidra

EDUCATION

B.S. Computer Science, University of Texas at San Antonio — May 2023

Concentration in Cyber Operations · Minor in Cybersecurity and Digital Forensics · GPA 3.5 · Honors

PUBLICATIONS

- “Predicting and Explaining Cognitive Load, Attention, and Working Memory in Virtual Multitasking.” IEEE, 2025.
- “Mazed and Confused: A Dataset of Cybersickness, Working Memory, Mental Load, Physical Load, and Attention During a Real Walking Task in VR.” 2024.

LEADERSHIP

Mentor, FIRST Tech Challenge · Family Service Association, San Antonio — Jul 2021 – Jul 2026

Mentored five students in engineering design and programming, leading the team to consistent regional qualifications, and helped secure $10,000 in robotics equipment for underserved communities.